Colin Duckworth

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Education

The University of Alabama, Tuscaloosa AL – BS in Electrical Engineering

Graduated May 2026

- GPA: 3.8/4.0 | 4x President's List, 7x Dean's List

Recent Projects

VHF/UHF Hydrology Radar Hardware Design *May 2026 - August 2026*

- RF lead for a team designing a 40–600 MHz FMCW radar payload for USGS river mapping, driving RF architecture through integration and deployment.
- Designed custom components & RFPCBs in Keysight ADS, fabricated layouts in house, and developed an AD8302-based reflection-cancellation circuit for shared single-antenna TX/RX operation.
- Engineered embedded control and power architecture (RFSoC + MCU) for RF-chain switching and LDO sequencing, integrating the 18 lb payload onto an in-house-built 40 lb-capacity UAS.
- Led RF chain integration, and the system was field-deployed in August 2026.

Ku/Ka-Band Glaciology Radar Front-End Design *August 2025 - May 2026*

- Developed the microwave TX/RX signal chain for a UAS-based polarimetric FMCW radar system (10 dBm dual-band, 12–18 GHz and 32–38 GHz) as part of a university research grant enabling remote sensing in arctic environments.
- Selected and integrated FMCW architecture components.
- Designed, fabricated, and characterized custom microwave PCBs for RF-chain integration and antenna design.

Underground Wireless Sensor Network for Mining Operations Capstone *August 2025 - April 2026*

- Contracted by Alpha Metallurgical Resources to design a wireless sensor network for real-time data transfer in underground mining environments for UA's new Senior Capstone initiative.
- Collaborated with Alpha's engineering team and university faculty to develop a low-latency (<50 ms), high-throughput (10 Mbps surge) wireless network that is platform agnostic - for future use in thermography and personnel tracking.
- Used Ansys HFSS to evaluate mine-environment signal propagation with various antennas, and used Sionna RT to perform ray tracing simulation to evaluate multipath reflections and channel responses within an underground mine.
- Designed a fiber-wireless network using point-to-point IEEE 802.11 at Layer 2, IPv4 static routing at Layer 3, and a fiber downlink from the control room to the head of the underground sensor daisy chain for reliable backhaul.

Work Experience

Graduate RF Engineering Intern, QuadSAT ApS – Odense Denmark *August 2026 – Current*

- Leading the front-end redesign of an electronic warfare (EW) payload for spectrum monitoring across the 1–18 GHz frequency range for a UAS/drone based platform, including implementation of a downconversion branch.
- Designed custom layouts and operating procedures integrating an MCU-tunable preselector to protect against jamming in congested and hostile spectrum environments.
- Transitioned the front end from connectorized components to a consolidated surface-mount layout, improving size, weight, and power (SWaP) performance.

RF Research Assistant, Remote Sensing Center – Tuscaloosa Alabama *August 2025 - August 2026*

- Owned end-to-end RF system design for FMCW radar payloads, from architecture and component selection through fabrication, integration, and field deployment for systems in UHF, Ku, and Ka bands.
- Extended scope beyond RF into embedded systems and power architecture to deliver complete UAS payloads.
- Applied ADS, HFSS, and COMSOL simulation alongside bench characterization to integrate increasingly complex payloads to meet sponsor requirements.
- Took an active technical role in the ongoing joint effort with the Alfred Wegener Institute.

Skills

Microwave/RF: Keysight ADS, Ansys HFSS, COMSOL Multiphysics, VNAs, Spectrum Analyzers

Hardware: Oscilloscopes, Impedance Analyzers, Function Generators, KiCad

Software: LTspice, MATLAB/Simulink, C/C++, Python